SOLUTIONS: CONTROL SYSTEMS – 2023/2024

1. Solution

a) Consider the three integrator blocks shown in the block scheme in Figure 1.1 in the text of the exam paper with outputs $v(t)$, $w(t)$, and $y(t)$ respectively. Clearly only these blocks represent dynamic systems with a state associated with. Therefore, a *possible choice* for the state variables is:

$$x_1(t) := v(t); \quad x_2(t) := w(t); \quad x_3(t) := y(t)$$

Hence, we define the state vector:

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

Thus, by inspecting the block scheme in Figure 1.1 in the text of the exam paper, we obtain:

$$\begin{cases} \dot{x}_1(t) = \dfrac{3}{2}u(t) \\ \dot{x}_2(t) = -\dfrac{4}{3}x_2(t) + 4u(t) \\ \dot{x}_3(t) = -x_3(t) + 5(x_1(t) + 2(-2x_3(t) + 3u(t) - x_2(t))) \\ y(t) = x_3(t) \end{cases}$$

After some trivial algebra, we get:

$$\begin{cases} \dot{x}_1(t) = \dfrac{3}{2}u(t) \\ \dot{x}_2(t) = -\dfrac{4}{3}x_2(t) + 4u(t) \\ \dot{x}_3(t) = 5x_1(t) - 10x_2(t) - 21x_3(t) + 30u(t) \\ y(t) = x_3(t) \end{cases}$$

Accordingly, the matrices A, B, C, D of the state space description are given by:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -4/3 & 0 \\ 5 & -10 & -21 \end{bmatrix}, \; B = \begin{bmatrix} 3/2 \\ 4 \\ 30 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \; D = 0$$

[15 Marks]

b) From the answer to Question 1-a), matrix A is given by

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -4/3 & 0 \\ 5 & -10 & -21 \end{bmatrix}$$

The matrix exponential e^{At} can be computed via the formula

$$e^{At} = \mathscr{L}^{-1}\left[(sI - A)^{-1}\right]$$

Answers

Now:

$$sI-A=\begin{bmatrix} s & 0 & 0 \\ 0 & s+4/3 & 0 \\ -5 & 10 & s+21 \end{bmatrix}$$

Thus:

$$\begin{aligned}(sI-A)^{-1} &= \frac{1}{s(s+4/3)(s+21)}\begin{bmatrix}(s+4/3)(s+21) & 0 & 5(s+4/3) \\ 0 & s(s+21) & -10s \\ 0 & 0 & s(s+4/3)\end{bmatrix}^{\top} \\ &= \frac{1}{s(s+4/3)(s+21)}\begin{bmatrix}(s+4/3)(s+21) & 0 & 0 \\ 0 & s(s+21) & 0 \\ 5(s+4/3) & -10s & s(s+4/3)\end{bmatrix} \\ &= \begin{bmatrix}\frac{1}{s} & 0 & 0 \\ 0 & \frac{1}{s+4/3} & 0 \\ \frac{5}{s(s+21)} & -\frac{10}{(s+4/3)(s+21)} & \frac{1}{s+21}\end{bmatrix}\end{aligned}$$

Finally, by proceeding with calculating the inverse Laplace transforms of each element of the matrix $(sI-A)^{-1}$ we obtain

$$\begin{aligned}e^{At} &= \mathscr{L}^{-1}\left[(sI-A)^{-1}\right] = \\ &= \begin{bmatrix}\mathbf{1}(t) & 0 & 0 \\ 0 & e^{-\frac{4}{3}t}\cdot\mathbf{1}(t) & 0 \\ \frac{5}{21}\left(1-e^{-21t}\right)\cdot\mathbf{1}(t) & \frac{30}{59}\left(e^{-21t}-e^{-\frac{4}{3}t}\right)\cdot\mathbf{1}(t) & e^{-21t}\cdot\mathbf{1}(t)\end{bmatrix}\end{aligned}$$

[15 Marks]

c) Owing to the answer to Question 1-a), we have

$$x_1(t) := v(t)\,; \quad x_2(t) := w(t)\,; \quad x_3(t) := y(t)$$

Moreover, from the text of Question 1-c), we have

$$v(0)=1\,; \quad w(0)=0\,; \quad y(0)=-1\,,$$

and thus

$$x(0) := \begin{bmatrix}1 \\ 0 \\ -1\end{bmatrix}$$

Therefore, after setting $u(t)=0, \forall t\in\mathbb{R}$, it follows that

$$\begin{aligned}x(t) &= e^{At}x(0) = \\ &= \begin{bmatrix}\mathbf{1}(t) & 0 & 0 \\ 0 & e^{-\frac{4}{3}t}\cdot\mathbf{1}(t) & 0 \\ \frac{5}{21}\left(1-e^{-21t}\right)\cdot\mathbf{1}(t) & \frac{30}{59}\left(e^{-21t}-e^{-\frac{4}{3}t}\right)\cdot\mathbf{1}(t) & e^{-21t}\cdot\mathbf{1}(t)\end{bmatrix}\cdot\begin{bmatrix}1 \\ 0 \\ -1\end{bmatrix}\end{aligned}$$

Answers

and

$$y(t) = Ce^{At}x(0) = x_3(t)$$

Easy algebra gives

$$x(t) = \begin{bmatrix} \mathbf{1}(t) \\ 0 \\ \left(\frac{5}{21} - \frac{26}{21}e^{-21t}\right) \cdot \mathbf{1}(t) \end{bmatrix}; \quad y(t) = \left(\frac{5}{21} - \frac{26}{21}e^{-21t}\right) \cdot \mathbf{1}(t)$$

[10 Marks]

Answers

2. Solution

a) Consider the block scheme shown in Figure 2.1 below which is the same as the one in Fig. 2.1 in Question 2 of the exam paper with the addition of the auxiliary variables $e(t)$, $v(t)$, $w(t)$, $q(t)$, $p(t)$, and $h(t)$.

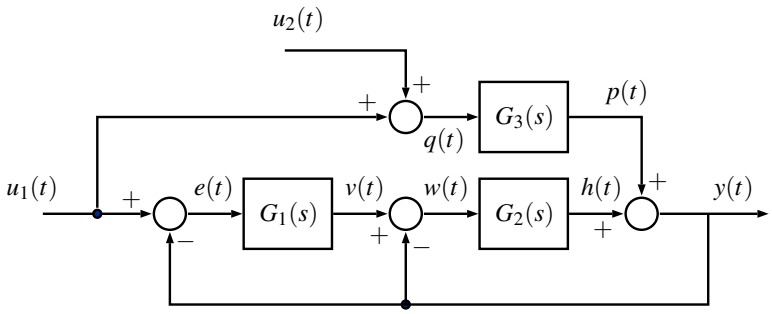


Figure 2.1 Block scheme for the solution of Question 2 with the addition of auxiliary variables.

By inspecting the block scheme in Figure 2.1 above, we immediately get the following relationships where, with some abuse of notation, we represent the inputs and outputs in the relations obtained from each block in the time-domain instead of in the Laplace domain:

$$h(t) = G_2(s)w(t)\,, v(t) = G_1(s)e(t)\,, q(t) = u_1(t) + u_2(t)\,, p(t) = G_3(s)q(t)\,,$$
$$w(t) = v(t) - y(t)\,, y(t) = h(t) + p(t)\,, e(t) = u_1(t) - y(t)$$

After some trivial algebra, by "eliminating" all the auxiliary variables and the associated relationships, we get:

$$y(t) = G_1(s)G_2(s)u_1(t) - G_1(s)G_2(s)y(t) - G_2(s)y(t) + G_3(s)u_1(t) + G_3(s)u_2(t)$$

Since we have to determine the transfer function $G_{u_1,y}(s)$ from the input $u_1(t)$ to the output $y(t)$, we set $u_2(t) = 0$ thus obtaining:

$$y(t) = G_1(s)G_2(s)u_1(t) - G_1(s)G_2(s)y(t) - G_2(s)y(t) + G_3(s)u_1(t)$$

Therefore:

$$G_{u_1,y}(s) = \frac{G_1(s)G_2(s) + G_3(s)}{1 + G_1(s)G_2(s) + G_2(s)}$$

Since:

$$G_1(s) = \frac{5}{s+5}; \quad G_2(s) = 12; \quad G_3(s) = \frac{2s}{1-5s}$$

after some easy calculations, it follows that:

$$G_{u_1,y}(s) = \frac{2s^2 - 290s + 60}{(1-5s)(13s+125)}$$

[10 Marks]

b) Let us recall from the answer to Question 2-a) the relationship:

$$y(t) = G_1(s)G_2(s)u_1(t) - G_1(s)G_2(s)y(t) - G_2(s)y(t) + G_3(s)u_1(t) + G_3(s)u_2(t)$$

Since we have to determine the transfer function $G_{u_2,y}(s)$ from the input $u_2(t)$ to the output $y(t)$, we now set $u_1(t) = 0$ thus obtaining:

$$y(t) = -G_1(s)G_2(s)y(t) - G_2(s)y(t) + G_3(s)u_2(t)$$

Answers

Therefore:

$$G_{u_2,y}(s) = \frac{G_3(s)}{1 + G_1(s)G_2(s) + G_2(s)}$$

Since:

$$G_1(s) = \frac{5}{s+5}; \quad G_2(s) = 12; \quad G_3(s) = \frac{2s}{1-5s}$$

after some easy calculations, it follows that:

$$G_{u_2,y}(s) = \frac{2s^2 + 10s}{(1-5s)(13s+125)}$$

[6 Marks]

c) Since the transfer functions $G_{u_1,y}(s)$ and $G_{u_2,y}(s)$ do not show any common factors between numerator and denominator and owing to the fact that the denominator is of degree 2, there is no hidden dynamics in the system represented by the block scheme in Fig. 2.1 of Question 2 of the exam paper.

The denominator of the transfer functions $G_{u_1,y}(s)$ and $G_{u_2,y}(s)$ shows one positive pole $p = 1/5$, and thus we conclude that the system represented by the block scheme in Fig. 2.1 of Question 2 of the exam paper is unstable.

It is worth noting that the instability of the whole system can also be ascertained by just inspecting the block scheme in Fig. 2.1 of Question 2 of the exam paper. In fact, the block represented by the transfer function $G_3(s)$ is "in parallel" with the rest of the system and hence its unstable pole $p = 1/5$ is also a pole of the whole system, thus causing its instability.

[4 Marks]

Answers

3. Solution

a) Requirement i) on the closed-loop steady-state static error $\bar{e}_\infty$ imposes the presence of an open-loop pole in $s=0$. Since this pole is not included in the poles of $G(s)$, it has to be added by the controller $C(s)$. Hence, we choose a controller of the form:

$$C(s) = C_1(s)\cdot C_2(s), \quad \text{with} \quad C_1(s) = \frac{\mu_C}{s} \text{ and } \quad C_2(s) = 1\,,$$

where $\mu_C > 0$ is a gain to be suitably designed. With the above choice of $C_1(s)$, Requirement i) is satisfied. The new open-loop transfer function denoted as $L_1(s)$ now reads:

$$L_1(s) = C_1(s)G(s) = \frac{\mu_C}{s}\cdot 12\cdot \frac{1+0.1\,s}{(1+20\,s)(1+3.3\,s)} = 12\mu_C\cdot \frac{1+0.1\,s}{s(1+20\,s)(1+3.3\,s)}$$

The parameters of

$$L_1(s) = 12\mu_C\cdot \frac{1+0.1\,s}{s(1+20\,s)(1+3.3\,s)}$$

are

$$\mu = 12\mu_C, \quad g = 1 \text{ (1 pole in } s=0),$$

and the relevant frequencies are:

$$\omega_{z_1} = 10, \quad \omega_{p_1} = 0, \quad \omega_{p_2} = \frac{1}{20}, \quad \omega_{p_3} = \frac{10}{33} \simeq 0.3.$$

To address the loop-shaping design to meet Requirements ii) and iii), we first set $\mu = 12\mu_C = 1$ and we sketch the Bode diagram straight-line approximation:

- When $0 < \omega < \frac{1}{20}$, the slope is $-20\ dB/dec$. Thus the magnitude is given by

$$\begin{aligned} |L_1|_{dB} &= -20\log\omega + \mu|_{dB} \\ &= -20\log\omega. \end{aligned}$$

For $\omega = \frac{1}{20} = 0.05$, the magnitude is

$$|L_1|_{dB} \simeq -20\log 0.05 \simeq 26dB.$$

- When $\frac{1}{20} \le \omega < \frac{10}{33}$, the slope is $-40dB/dec$. Thus the magnitude is given by

$$\begin{aligned} |L_1|_{dB} &= -40(\log\omega - \log 0.05) + 26 \\ &\simeq -40\log\omega - 26. \end{aligned}$$

For $\omega = \frac{10}{33}$, the magnitude is

$$|L_1|_{dB} \simeq -40\log\frac{10}{33} - 26 \simeq -5.28dB.$$

- When $\frac{10}{33} \le \omega < 10$, the slope is $-60dB/dec$. Thus the magnitude is given by

$$\begin{aligned} |L_1|_{dB} &= -60(\log\omega - \log\frac{10}{33}) - 5.28 \\ &\simeq -60\log\omega - 36.39. \end{aligned}$$

For $\omega = 10$, the magnitude is

$$|L_1|_{dB} \simeq -60\log 10 - 36.39 \simeq -96.4dB.$$

Answers

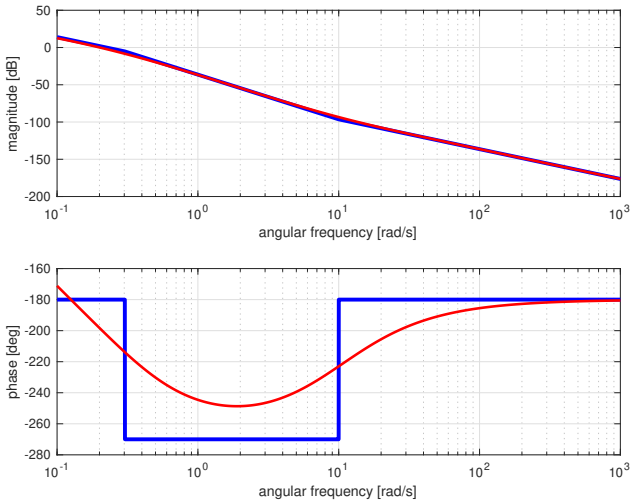
- When $\omega \geq 10$, the slope is $-40dB/dec$. Thus the magnitude is given by

$$\begin{aligned} |L_1|_{dB} &= -40(\log\omega - \log 10) - 96.4 \\ &\simeq -40\log\omega - 56.39. \end{aligned}$$

For $\omega = 20$, the magnitude is

$$|L_1|_{dB} \simeq -40\log 20 - 56.39 \simeq -108.4dB.$$

The resulting true Bode diagram with its straight-line approximation is plotted in figure below.



By inspecting the Bode diagrams in figure above, it turns out that $\arg L_1(j\omega) < -180°, \forall\,\omega$ such that $5 \leq \omega \leq 15$. Hence, there does not exist a value of the controller gain μ_C making the closed-loop system asymptotically stable while, at the same time, satisfying Requirement ii).

Therefore, a significant increase of $\arg L_1(j\omega) < -180°, \forall\,\omega$ such that $5 \leq \omega \leq 15$ is needed. It is worth noting that increasing $\arg L_1(j\omega)$ would also be beneficial in terms of meeting Requirement vi).

Since $G(s)$ is asymptotically stable and minimum phase, introducing a dynamic controller that "cancels" some or all the open-loop poles and zeros is permitted.

Therefore, we select the following dynamic controller:

$$C_2(s) = \frac{(1+20s)(1+3.3s)}{1+0.1s}$$

thus getting

$$C(s) = C_1(s)\cdot C_2(s) = \mu_C \frac{(1+20s)(1+3.3s)}{s(1+0.1s)}$$

hence obtaining the open-loop transfer function

$$L_2(s) = 12\mu_C \frac{1}{s}$$

The phase of the open-loop transfer function now reads:

$$\arg L_2(j\omega) = -90°\,,\ \forall\,\omega > 0$$

Answers

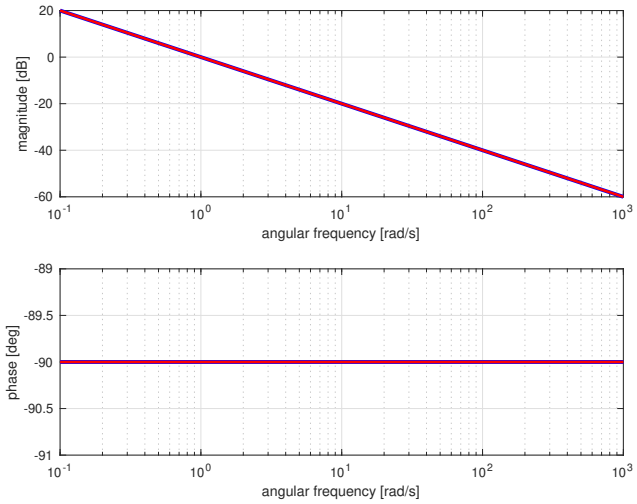
As a consequence, with the above selection of the controller $C(s)$, the closed-loop system is asymptotically stable and Requirements i) and iv) are simultaneously satisfied.

Again, we set $\mu = 12\mu_C = 1$ and we sketch the Bode diagrams with the new choice of the controller $C(s)$.

Since

$$L_2(s) = 12\mu_C \frac{1}{s} = \mu \frac{1}{s} = \frac{1}{s}$$

the straight-line approximation coincides with the true Bode diagrams shown in figure below.



By inspecting the above Bode diagrams, the slope of $|L_2(j\omega)|_{dB}$ is constant and equal to $-20dB/dec$ and $\arg L_2(j\omega) = -90°$. Therefore, the closed-loop pole is given by $p_{\text{cl}} = -1/\omega_c$, where ω_c is the crossover frequency, and, accordingly, the settling-time at 1% is given by the approximate formula:

$$t_{s,0.01} \simeq \frac{4.6}{\omega_c}$$

Therefore, by choosing $\omega_c = 10\text{rad/sec}$ (which satisfies Requirement ii)) we obtain

$$t_{s,0.01} \simeq \frac{4.6}{\omega_c} = \frac{4.6}{10} = 0.46 < 2\,\text{sec}$$

and thus Requirement iii) is met, too.

Finally, to get $\omega_c = 10\text{rad/sec}$, we select the gain μ_C as follows:

$$-20\log 10 + 20\log 12\mu_C = 0 \Longrightarrow \log 12\mu_C = 1 \Longrightarrow \mu_C = \frac{10}{12} = \frac{5}{6}$$

thus concluding the design of the controller $C(s)$ which is given by

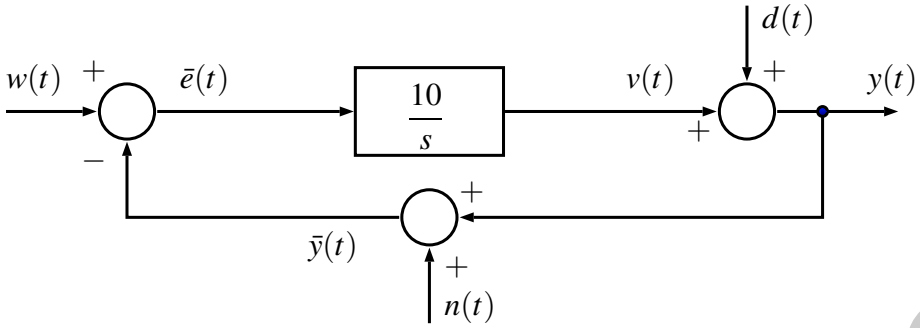
$$C(s) = \frac{5}{6}\frac{(1+20s)(1+3.3s)}{s(1+0.1s)}$$

Remark for the students. To answer this question, using only the straight-line approximations of the Bode diagrams is fully acceptable to get full marks.

[20 Marks]

ANSWERS

b) Consider the block scheme shown in figure below that is obtained from the block scheme given in Question 3 in the text of the exam paper using the controller $C(s)$ obtained in the answer to Question 3-a).



The Laplace transforms of the inputs $w(t)$, $d(t)$, and $n(t)$ are given by:

$$\mathscr{L}[w(t)] = \frac{3}{s}; \quad \mathscr{L}[d(t)] = \frac{2}{s}e^{-5s}; \quad \mathscr{L}[n(t)] = \frac{0.5}{s}e^{-s}$$

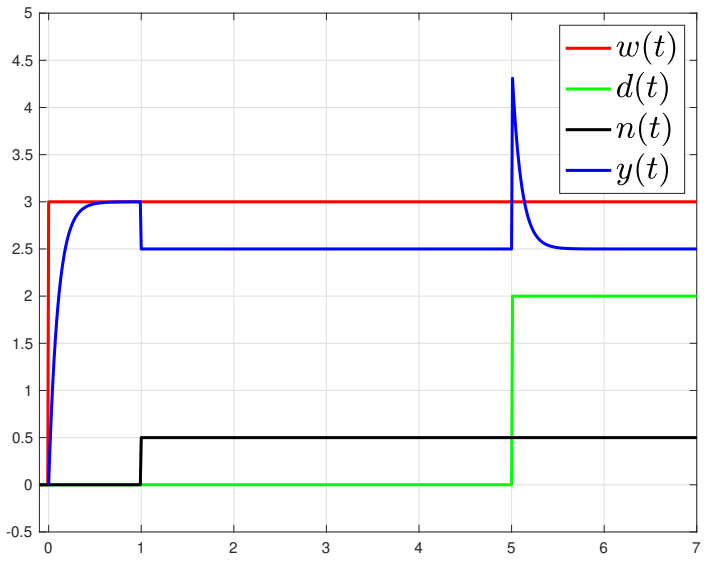
By linearity arguments and easy calculations of the transfer functions from the inputs $w(t)$, $d(t)$, and $n(t)$ to the output $y(t)$, the Laplace transform of the output $y(t)$ can be obtained as follows:

$$Y(s) = \mathscr{L}[y(t)] = \frac{10}{s+10}\cdot\frac{3}{s} + \frac{s}{s+10}\cdot\frac{2}{s}e^{-5s} - \frac{10}{s+10}\cdot\frac{0.5}{s}e^{-s}$$

By suitable partial fraction expansions we immediately obtain

$$y(t) = \mathscr{L}^{-1}[Y(s)] = 3\left(1-e^{-10t}\right)\cdot\mathbf{1}(t) + 2e^{-10(t-5)}\cdot\mathbf{1}(t-5) - 0.5\left(1-e^{-10(t-1)}\right)\cdot\mathbf{1}(t-1)$$

In the following figure, the time-behaviours of the input and output variables are shown.



The overshoot that can be noticed at time $t = 5sec$ is caused by the action of the disturbance $d(t)$. Since Requirement iv) in Question 3-a) is specified for $d(t) = n(t) = 0$, no overshoot would occur in such a case, thus confirming

Answers

the correctness of the designed controller $C(s)$. In the figure, the effects of the disturbances $d(t)$ and $n(t)$ are shown as well. It can be observed that the effect of $d(t)$ is vanishing asymptotically due to the presence of the integrator in the controller $C(s)$. Also, it can be observed that the effect of $n(t)$ instead is equivalent to a change in the reference input $w(t)$, which is in accordance with the theory: the controller $C(s)$ cannot mitigate the effects of a disturbance on the feedback path when this disturbance is acting in the same frequency range of the reference input $w(t)$.

[8 Marks]

c) Due to the presence of the integrator in the controller $C(s)$, the effect of the disturbance $d(t) = 5 \cdot \mathbf{1}(t)$ is asymptotically vanishing. Therefore, the input $d(t)$ can be neglected when considering the calculation of the steady-state output $y_\infty(t)$.

Hence, it suffices to consider the effect of the input $n(t) = 2 \cdot \sin(\overline{\omega} t) \cdot \mathbf{1}(t)$ through the transfer function $\widetilde{F}(s)$ from the input $n(t)$ to the output $y(t)$ which is given by

$$\widetilde{F}(s) = \frac{\mathscr{L}[y(t)]}{\mathscr{L}[n(t)]} = -\frac{10}{s+10}$$

Since the closed-loop system is asymptotically stable, the magnitude $|y_\infty(t)|$ of the steady-state output generated by the input $n(t) = 2 \cdot \sin(\overline{\omega} t) \cdot \mathbf{1}(t)$ can be evaluated using the frequency response theorem:

$$|y_\infty(t)| = 2 \cdot \left|\widetilde{F}(j\overline{\omega})\right| = 2 \cdot \left|\frac{10}{j\overline{\omega}+10}\right|$$

Easy calculations show that

$$\overline{\omega} > 20 rad/sec \Longrightarrow 2 \cdot \left|\frac{10}{j\overline{\omega}+10}\right| < 0.1$$

Thus, the high-frequency range $[\omega^-, +\infty)$ such that

$$|y_\infty(t)| < 10^{-1}, \quad \forall \overline{\omega} \in [\omega^-, +\infty)$$

is given by $[20 rad/sec, +\infty)$.

[12 Marks]

Answers